# SAHO – Product Requirements Document

## Phase 6 – Database Design & API Specification

## 6.1 Objective

Define the backend data model, Firestore collections, API contracts, authentication flow, and service interfaces for SAHO. The design emphasizes scalability, privacy, and maintainability.

## 6.2 Firestore Collections

| Collection | Purpose |
| --- | --- |
| users | User profile and preferences |
| caregiver_contacts | Trusted emergency contacts |
| recovery_sessions | AI recovery interactions |
| timeline_entries | Recovery milestones and reflections |
| education_progress | Learning history |
| app_settings | Accessibility and notification preferences |
| audit_logs | Operational events (non-sensitive) |

## 6.3 Core Data Models

User - id - displayName - email - recoveryGoals - accessibilityPreferences RecoverySession - sessionId - userId - timestamp - emotion - riskLevel - aiActions - emergencyTriggered CaregiverContact - contactId - name - relationship - phone - emergencyEnabled

## 6.4 Authentication Flow

Firebase Authentication supports: • Anonymous guest access for emergency features • Email/password • Google Sign-In Only authenticated users can synchronize recovery history across devices.

## 6.5 API Service Layer

Client components communicate only with services. Services: • AuthService • AIService • RecoveryService • ContactService • VisionService • EducationService • TimelineService

## 6.6 API Contracts

POST /api/ai/pulse Input: emotion, transcript Output: structured JSON guidance POST /api/vision/analyze Input: image Output: trigger assessment or substance identification GET /api/timeline Returns recovery history. POST /api/caregiver/notify Sends emergency notifications after user confirmation.

## 6.7 Validation

All requests must be validated using Zod. Invalid requests return structured error responses. Images must be checked for file type and size before processing.

## 6.8 Security Rules

Firestore Rules: • Users access only their own documents. • Caregiver records require owner permission. • Audit logs are write-only from backend. • No public read access to sensitive collections.

## 6.9 Offline Synchronization

Recovery sessions created offline are stored locally and synchronized when connectivity returns. Conflicts are resolved using server timestamps.

## 6.10 Acceptance Criteria

The backend architecture is approved when: • Firestore collections are normalized. • APIs return typed responses. • Authentication protects user data. • Offline synchronization works reliably. • Validation prevents malformed requests. • Security rules pass automated tests.